Your first plate count

Install the Windows application, learn the main controls, then practise on the two synthetic plates.

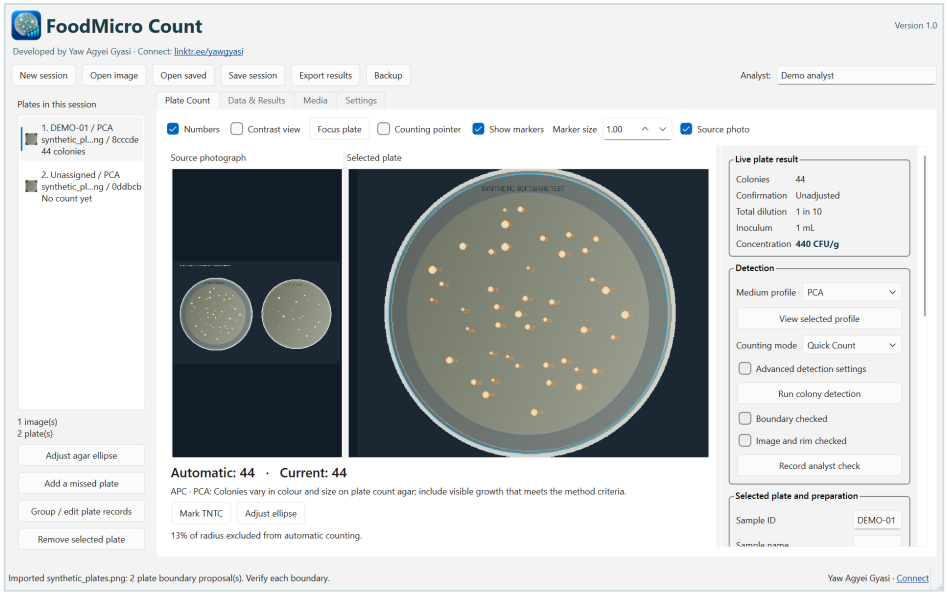
Where to open the app

Search **FoodMicro Count** in Windows Start. The same Start menu group includes **FoodMicro Count Beginner Guide**.

1. Install. Run the Windows x64 Setup. Read the pre-installation notice, review the licence, choose the program folder, and read the first-run page after installation.

2. Find the practice image. In File Explorer, paste **%LOCALAPPDATA%\Programs\FoodMicro Count\demo** in the address bar. Use the folder selected during Setup if you changed the location. The file is **synthetic_plates.png**.

3. Enter an analyst name. Use the box at the top right. The name is recorded with corrections and review.



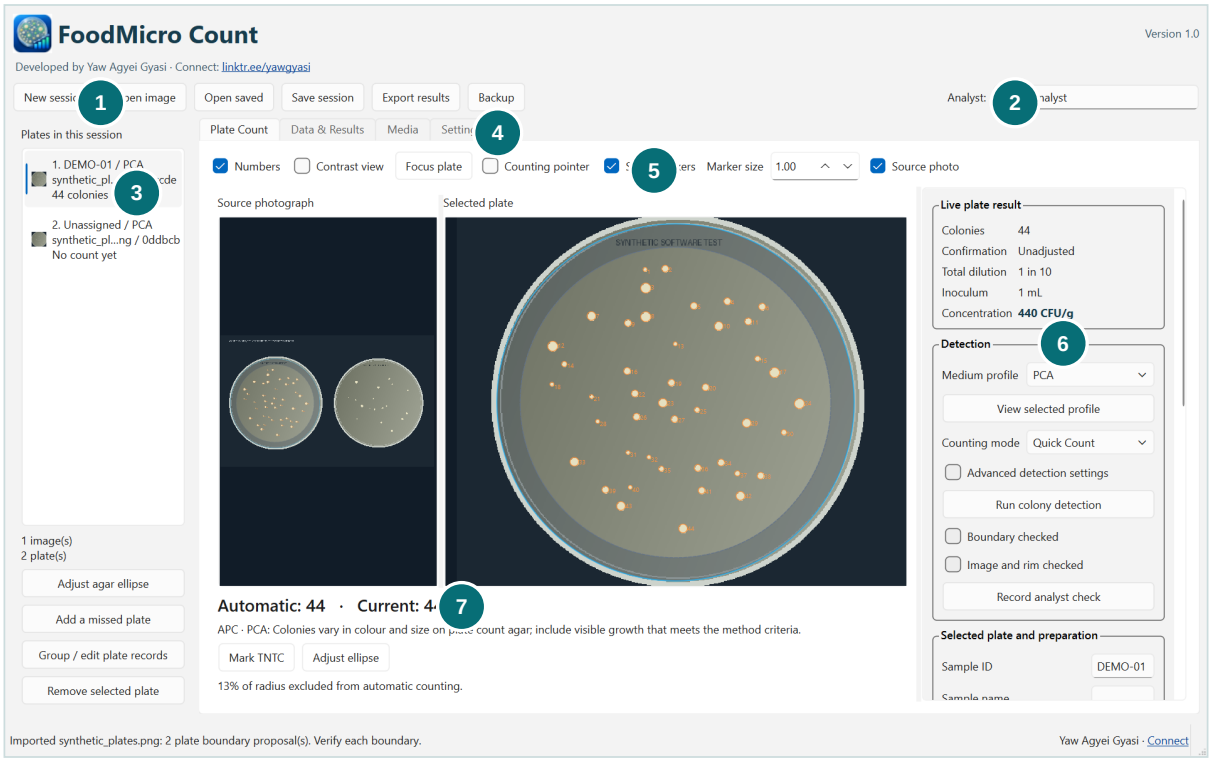
The installed Windows interface with a synthetic plate and automatic colony markers.

Before working with laboratory data

The exercises use generated images. Review every boundary, marker, observation state and method before using a result. Automatic counts vary with image scale and acquisition conditions.

Find the right button

The four tabs separate image counting, calculations, media presets and local settings.



Numbers locate the main control groups. The selected dish stays visible while you review the count.

1 Session toolbar. New session starts a separate session. **Open image** imports a photo, **Open saved** reloads a session, and **Save session** records a revision. **Export results** and **Backup** create copies for different purposes.

2 Analyst. Enter the name to record with corrections and review.

3 Plate list. Select one dish. Use the buttons below it to adjust, add, group or remove a plate.

4 Tabs. Plate Count is the image workspace. **Data & Results** shows calculations. **Media** manages presets. **Settings** shows the local folder and diagnostics.

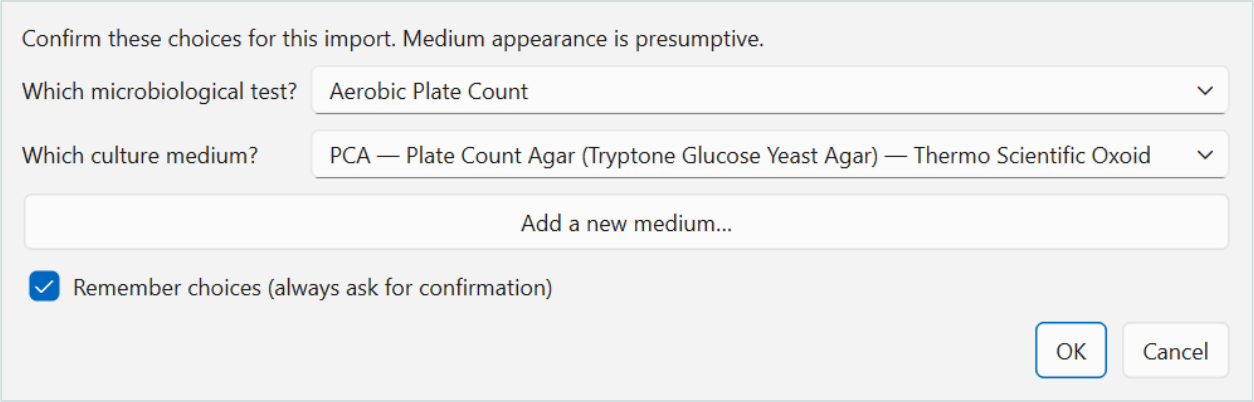
5 View tools. Toggle numbers, contrast, markers and source photo. **Focus plate** opens full screen.

6 Detection and record. Choose the medium and count mode. Scroll this panel to reach preparation and reporting fields.

7 Count summary. Compare automatic and current totals. Use **Mark TNTC** for an appropriate non-numeric observation.

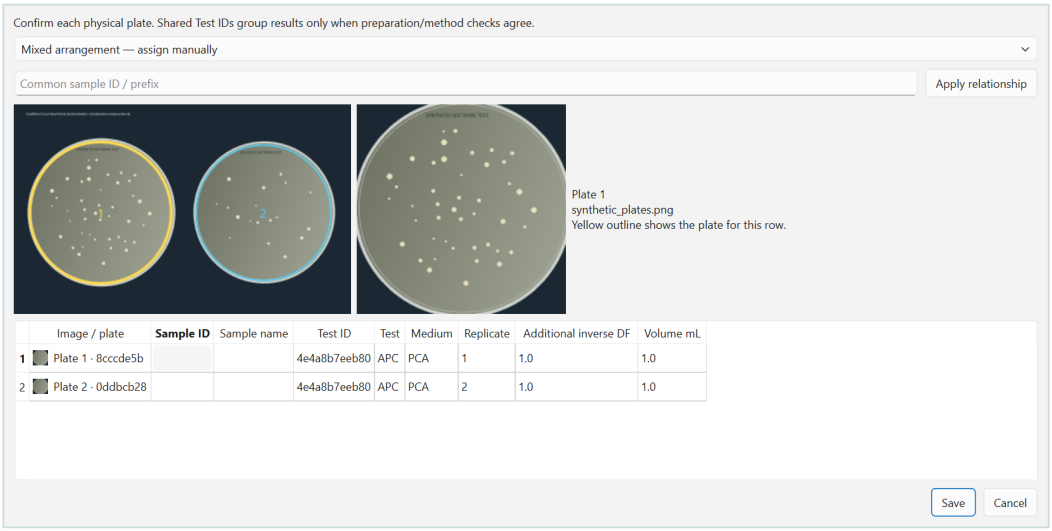
Open a plate photograph

Bring in the image, identify each dish, and inspect the agar outline before counting.



Import setup asks for the test and medium. The app also accepts drag-and-drop from File Explorer.

Choose the test and medium. For the practice image use **Aerobic Plate Count** and **PCA**. Select the actual test and medium for a laboratory plate. For a mixed batch, configure images separately.



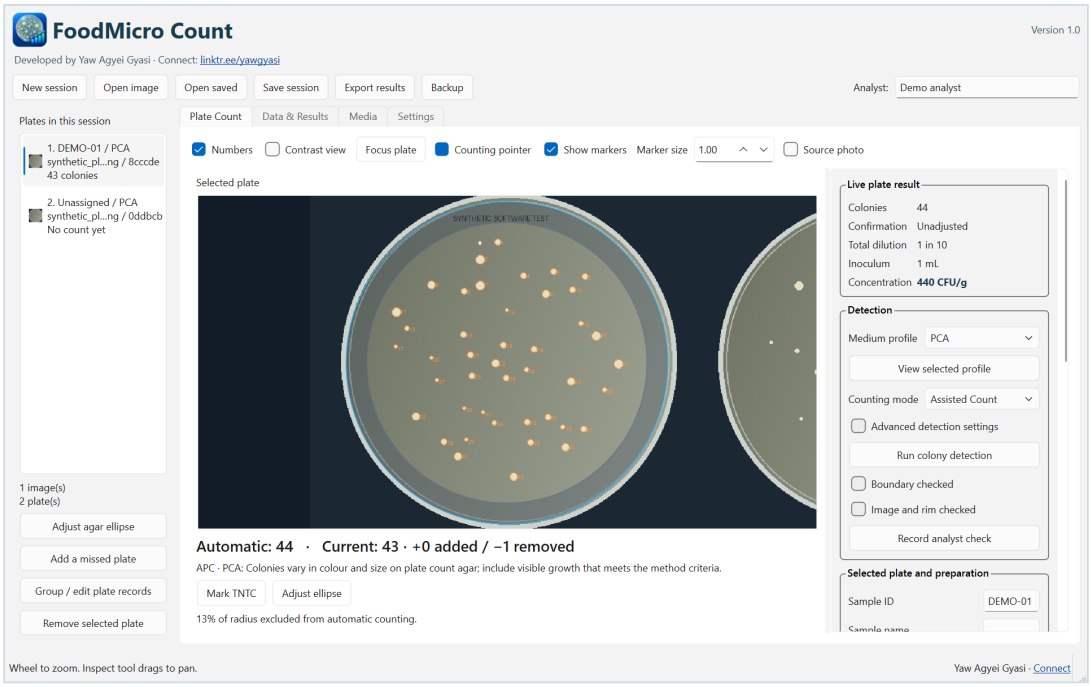
The grouping dialog lists plate proposals. Give each plate the sample, test and replicate identity that applies.

Check the outline. Select a dish and click **Adjust agar ellipse** or **Adjust ellipse**. Drag to draw an ellipse or use the centre, edge and rotation handles. Use **Add a missed plate** for a dish the app did not propose.

Inspect the source. Scroll to zoom and middle-drag to pan. Right-click and choose **Fit image to window** to reset the view. F11 opens full screen, and Esc returns.

Turn markers into a count

Choose a counting mode, then check the selected dish against its original image.



Practice correction. Automatic stays at 44. Removing one included marker changes Current to 43.

Quick Count. Choose the mode and click **Run colony detection**. Check the orange automatic markers against the image.

Assisted Count. Run detection with the counting pointer ready. Click a missed centre inside the ellipse to add. Shift+click a marker to remove. Ctrl+click selects markers for merging.

Manual Count. Starts a separate empty manual layer and retains the previous layer. Click only centres you can distinguish. Use right-click **Count history** to retain or restore a layer.

Correct a cluster or artefact. Right-click the marker for **Split touching colonies**, **Include marker** or **Exclude as ink or artefact**. To merge, Ctrl+click the markers and select **Merge selected markers**.

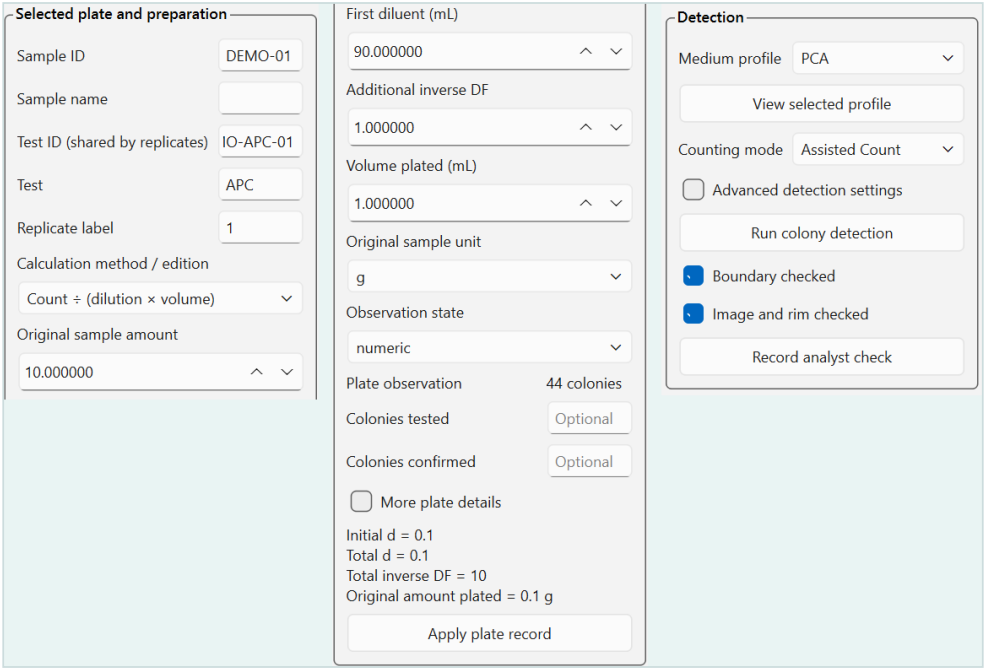
Undo or redo. Ctrl+Z reverses a correction and Ctrl+Y reinstates it. Use **Advanced detection settings** for sensitivity, radius, colour, watershed and rim controls.

Inspect difficult regions

Check glare, writing, edge growth and touching colonies at full resolution. If growth cannot be counted reliably, record the appropriate observation state.

Record what was plated

The sample, dilution and volume define how a plate count is converted to a concentration.



From left to right, enter the sample and method, finish the preparation fields, then record the analyst check.

Identify the observation. Enter the Sample ID, Test ID and test. Replicates of one sample and test share a Test ID and use distinct replicate labels. Use different Test IDs for independent observations.

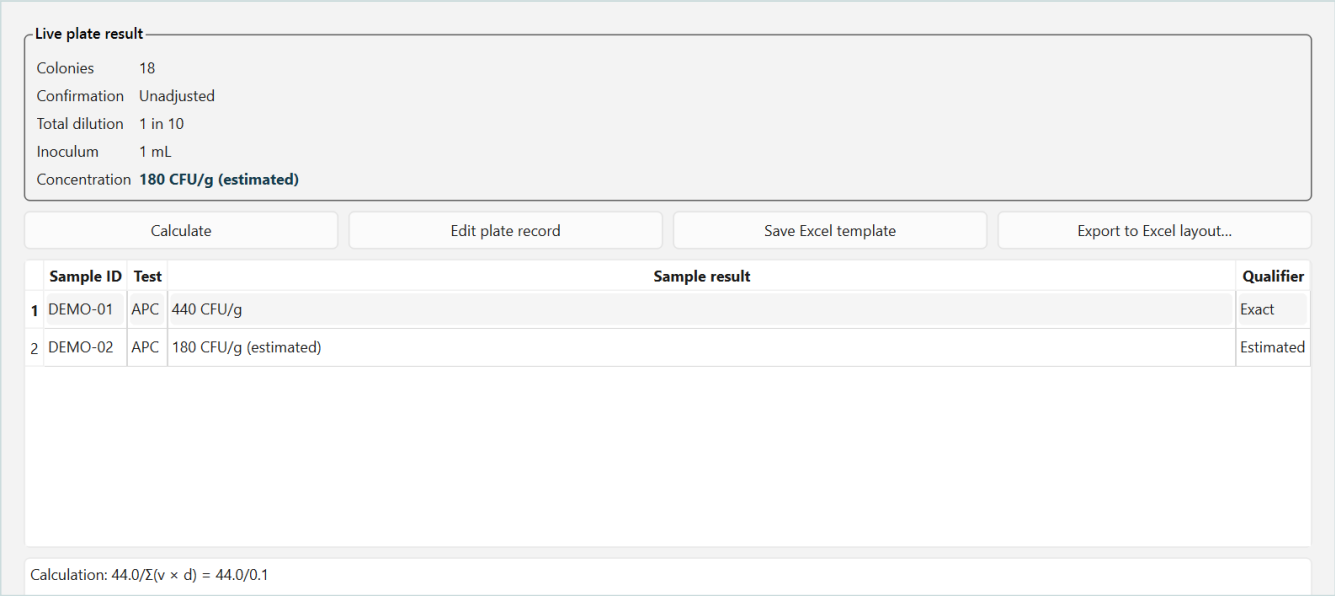
Enter the preparation. Enter sample amount, first diluent, original unit and plated volume. **Additional inverse DF** is the reciprocal factor after the initial preparation. With 10 g plus 90 mL, enter 1 when there is no further dilution.

Check the method and state. Select the intended method. Use **numeric** for an enumerable plate. Choose the matching state for zero, spreading, contamination, unreadable, missing or other observations. Confirmatory counts need both tested and confirmed values.

Record analyst review. Click **Apply plate record**. Verify the boundary and image, tick **Boundary checked** and **Image and rim checked**, then click **Record analyst check**. Review again after editing.

Read, save and export

Check the number, qualifier, unit, input plates and calculation trace before sharing results.



Data & Results displays two separately identified synthetic practice samples.

Calculate or edit. Calculate refreshes results. Select a row to see the trace. **Edit plate record** returns to the plate inputs. An estimated value or reporting bound must travel with its qualifier.

Save and reopen. Save session creates a local revision. **Open saved** reloads a selected session. Use **Backup** to copy the database, originals and local presets to a separate folder.

Export. Export results creates Excel, CSV, JSON and annotated PNGs. In Excel, begin with **Sample results**, then check **Plate Observations** and **Calculation Details**. Keep the traceability JSON with the records.

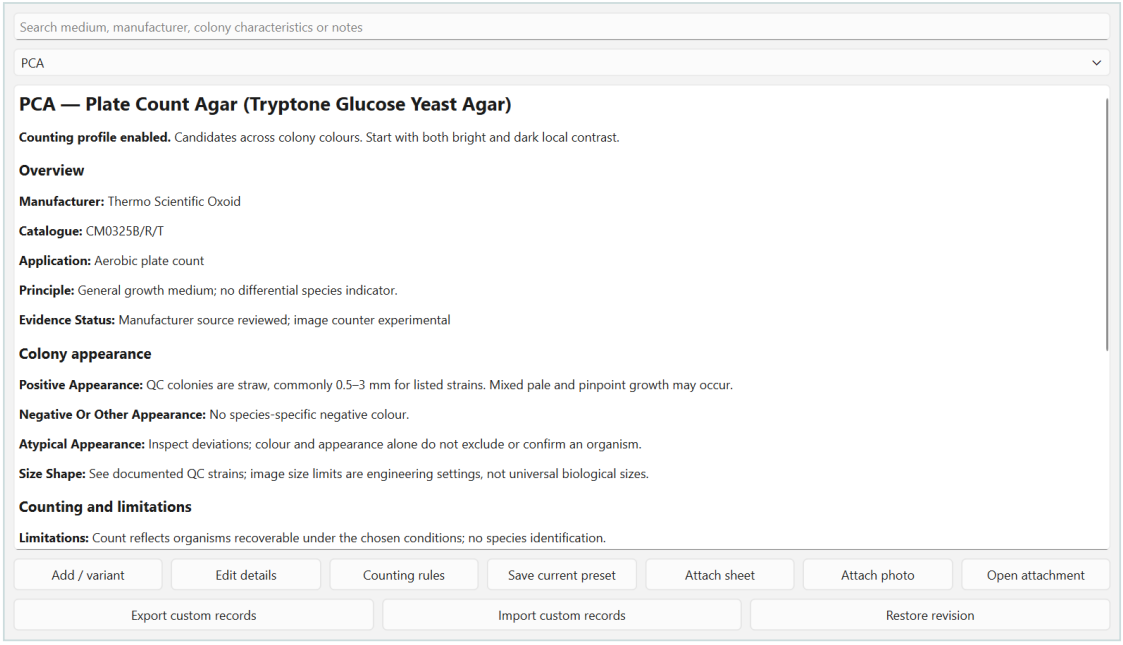
Import a spreadsheet. In **Data & Results**, open **Spreadsheet import and manual data entry**. Choose **Import Excel**, confirm the worksheet and column mappings, then **Review selected rows**. Excel records stay separate from image counts.

Understand missing and zero

Zero means no colonies were observed under the documented conditions. A missing record and a less-than reporting bound are different observations.

Choose and manage presets

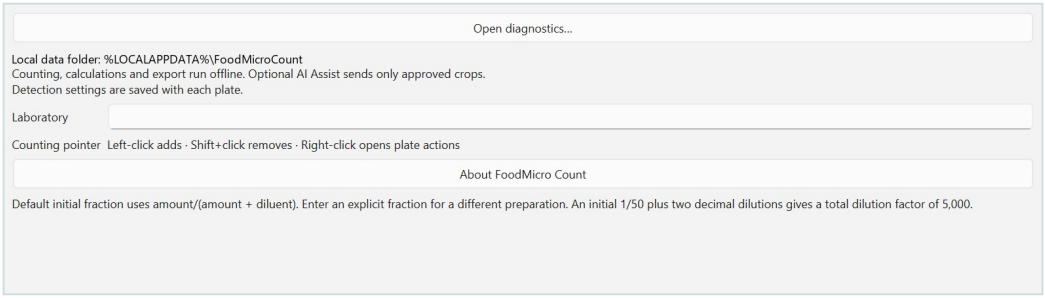
Use the medium library to read appearance guidance and choose counting settings for the current task.



The Media tab shows the selected medium, source guidance and editable presets.

Browse and select. Search the medium in **Media**. Use **View selected profile** in Plate Count to jump to the current entry. Appearance guides help you recognise candidates for review.

Save or restore. Use **Counting rules** to edit permitted settings and **Save current preset** to keep a local preset. Use **Restore revision** and choose **Bundled counting preset** to restore the built-in detector settings.



Settings gives the local data location, diagnostics and the About dialog.

Check local settings. Open diagnostics shows the current plate warnings. Optional AI Assist sends selected crops after you approve the preview. Counting and export work offline.

Practise on two dishes

Both examples use PCA and the Count ÷ (dilution × volume) method in the preparation panel.

Sample A / 44 marks. Select the first dish, enter Sample ID **DEMO-01** and Test ID **DEMO-APC-01**, then run detection. Enter 10 g, 90 mL first diluent, additional inverse DF 1, and 1 mL plated. Review the plate and open **Data & Results**.

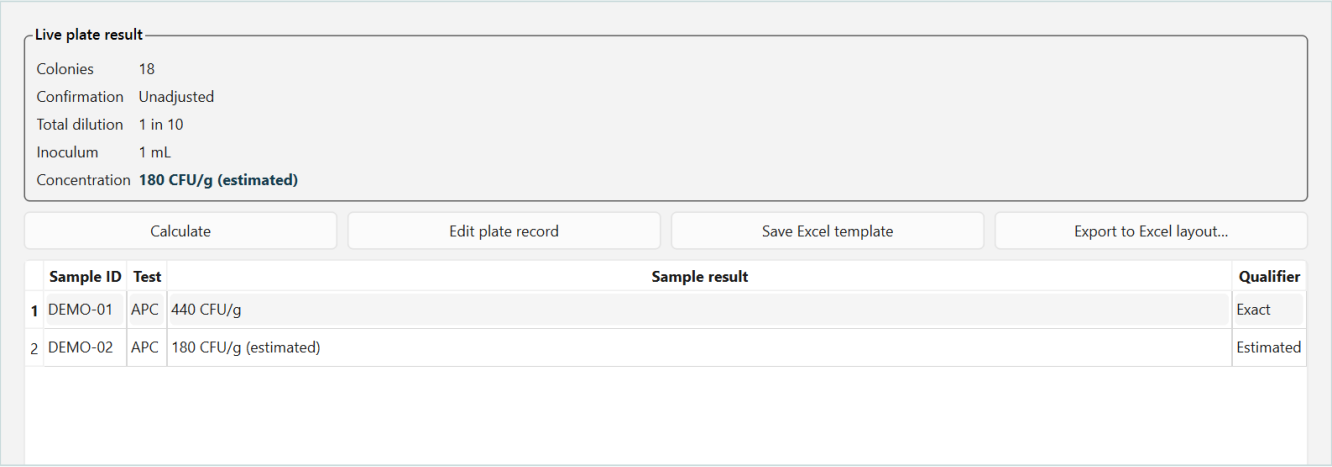
Expected arithmetic for Sample A

44 / (1 mL × 0.1) = 440 CFU/g. Initial fraction is 10 / (10 + 90) = 0.1. Keep the input volume and dilution in the trace.

Sample B / 18 marks. Select the second dish, assign Sample ID **DEMO-02** and Test ID **DEMO-APC-02**, and use the same preparation values. Treat it as a separate sample for this exercise. Run detection, review it and check the qualifier.

Expected arithmetic for Sample B

18 / (1 mL × 0.1) = 180 CFU/g (estimated). The count is below the basic method's configured 25-colony lower limit. Preserve the estimated qualifier.



The reviewed practice session shows 440 CFU/g and 180 CFU/g (estimated).

Try a correction. On Sample A, Shift+click one included marker. Confirm that Current falls to 43 while Automatic stays at 44. Press Ctrl+Z and confirm that Current returns to 44. Record the analyst check again before saving.

Try a manual session

Use a separate training session to practise marking, recovery and choosing an observation state.

Exercise 1 / manual. Choose **New session** and open the same demo. Select one dish, choose **Manual Count** and mark five clear centres. Check that the current total is 5. Remove one with Shift+click, then undo. This partial exercise practises the controls. Leave it in a training session.

Exercise 2 / spreadsheet. Use **Save Excel template**. Enter a practice observation with a distinct sample and test ID, count 44, total inverse dilution 10 and volume 1 mL. Import the workbook, review the row and compare the result with Sample A if you chose the same method and unit.

Exercise 3 / save and recover. Save the practice session. Open it again through **Open saved**. Create a backup and note its location. Check the backup folder before relying on it.

When a result is blank

Check Sample ID, Test ID, volume, observation state, calculation method and analyst review. Open the calculation trace for the reason. ISO 7218:2024 calculation is blocked pending controlled method verification.

When a count looks wrong

Check the selected dish, medium and agar ellipse. Inspect glare, writing, shadows and overlapping growth. Adjust only the settings you understand, or use manual review. Record TNTC or another applicable state when an exact count cannot be justified.

Where to find files

The program folder contains the PDF guide, synthetic demo, licence and source archive. The separate records folder is shown in **Settings**.